

CO1 Assessment Tool-2

CSA1007-Software Engineering

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Problem Statement: Intelligent Nuclear Power Plant Monitoring Platform

1. Project Vision and Stakeholders

Project Vision

Develop an Intelligent Nuclear Power Plant Monitoring Platform that continuously monitors reactor conditions, detects operational anomalies, predicts equipment failures, tracks radiation levels, supports emergency response, and generates compliance analytics to improve plant safety, reduce operational risks, ensure regulatory compliance, minimize downtime, and support better decision-making.

Project Objectives:

- Monitor reactor conditions.
- Detect operational anomalies.
- Predict equipment failures.
- Track radiation levels.
- Support emergency response.
- Generate compliance analytics.

Stakeholders:

- Plant Operators – Monitor reactor status and plant operations.
- Control Room Engineers – Analyze system data and respond to abnormal conditions.
- Maintenance Engineers – Perform preventive and corrective maintenance.
- Safety Officers – Monitor radiation levels and ensure plant safety.
- Plant Administrator – Manage users, reports, and overall system.
- Regulatory Authorities – Review compliance reports and safety records.
- Emergency Response Team – Handle emergency situations using real-time information.

2. Create Epics and User Stories

Epic 1: Reactor Monitoring

As a Plant Operator, I want to monitor reactor conditions in real time, so that I can ensure safe plant operation.

Epic 2: Operational Anomaly Detection

As a Control Room Engineer, I want the system to detect operational anomalies automatically, so that I can take immediate corrective actions.

Epic 3: Equipment Failure Prediction

As a Maintenance Engineer, I want the platform to predict equipment failures, so that maintenance can be scheduled before breakdowns occur.

Epic 4: Radiation Monitoring

As a Safety Officer, I want to track radiation levels continuously, so that worker and environmental safety can be maintained.

Epic 5: Emergency Response

As an Emergency Response Officer, I want immediate alerts during critical situations, so that emergency actions can be initiated quickly.

Epic 6: Compliance Analytics

As a Plant Administrator, I want compliance reports to be generated automatically, so that regulatory requirements are satisfied.

3. Define Acceptance Criteria

User Story 1

Given reactor sensors are active

When the operator opens the dashboard

Then current reactor conditions shall be displayed in real time.

User Story 2

Given abnormal operating values occur

When the system detects anomalies

Then an alert shall be generated immediately.

User Story 3

Given equipment performance data is available

When failure probability exceeds the threshold

Then predictive maintenance recommendations shall be displayed.

User Story 4

Given radiation sensors are functioning

When radiation exceeds the safe limit

Then the system shall notify safety officers immediately.

User Story 5

Given a critical emergency is detected

When emergency conditions occur

Then alerts and emergency response procedures shall be activated.

User Story 6

Given monitoring data is stored

When compliance reports are requested

Then the system shall generate regulatory compliance reports automatically.

4. Prioritize the Product Backlog

High: Monitor reactor conditions (Very High Business Value)

High: Detect operational anomalies (Very High Business Value)

High: Track radiation levels (Very High Business Value)

High: Emergency response alerts (Very High Business Value)

Medium: Predict equipment failures (High Business Value)

Medium: Compliance analytics (Medium Business Value)

5. Plan the Sprint Backlog

- Monitor reactor conditions – Design dashboard, integrate sensor data, test monitoring.
- Detect operational anomalies – Develop anomaly detection module, create alerts, testing.
- Track radiation levels – Integrate sensors, display dashboard, generate alerts.
- Emergency response – Implement emergency notification module and test alert mechanism.

6. Estimate Effort Using Story Points

Estimation Technique: Fibonacci Sequence

- Monitor reactor conditions – 8 Story Points
- Detect operational anomalies – 8 Story Points
- Predict equipment failures – 13 Story Points
- Track radiation levels – 5 Story Points
- Emergency response – 5 Story Points
- Compliance analytics – 3 Story Points

7. Sprint Goal and Expected Deliverables

Sprint Goal:

Develop the first version of the Intelligent Nuclear Power Plant Monitoring Platform capable of monitoring reactor conditions, detecting operational anomalies, tracking radiation levels, and generating emergency alerts.

Expected Deliverables:

- Reactor Monitoring Dashboard
- Operational Anomaly Detection Module
- Radiation Monitoring Module
- Emergency Alert System
- Initial Product Backlog
- Sprint Backlog
- Story Point Estimation
- Sprint Documentation